Who Influenced the Decision?

Attribution in LLM Multi-Agent Discussions: State of the Art, Formalization and Plan

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Abstract

We study small groups of LLM agents (2 to 4) that discuss a multiple-choice question over a few rounds before a collective decision. Our goal is to explain that decision: which sentence, written by which agent, caused it. Attribution methods designed for a single model only see what the *last* agents read. In a discussion, however, information often reaches the decision *through* another agent who restates it, and these methods then credit the agent who repeated the idea rather than the one who brought it. We call this *attribution laundering*. This note (i) summarizes the state of the art, (ii) formalizes the setting and two ways of measuring influence (frozen vs. re-run deletion), (iii) proposes a cheap estimator that composes local attributions along the discussion, and (iv) describes how we will validate it against a ground truth.

1 State of the art

Small-scale multi-agent collaboration. In *multi-agent debate*, several LLMs answer independently, read each other's answers, revise them over a few rounds, and then aggregate them, usually by majority vote [4]. ReConcile [1] runs a round table between different LLMs with a confidence-weighted vote and improves reasoning on seven benchmarks (StrategyQA, GSM8K, ...); its ablations show that model diversity matters. Later work tempers these gains. Most of the improvement of debate is explained by majority voting alone [2]. Agents often change their answer by conformity rather than persuasion [5]. Groups fail to pool information that is distributed among agents [7]. All these studies are *behavioural*: they measure outcomes, not what caused a given decision.

Attribution for a single model. Attribution methods score the parts of an input by their influence on a model's output. **ContextCite** [3] splits the context into sentences. It deletes random subsets of them (about 32 times) and re-reads the probability of the original answer each time. It then fits a sparse linear model that predicts this probability from which sentences were present; the weight of each sentence is its score. **Interpreto** [10] implements perturbation methods (occlusion, LIME, KernelSHAP) and gradient methods (Integrated Gradients, saliency) at the token, word or sentence level, as well as concept-based explanations. All of these explain *one model reading one fixed input*: nothing is re-generated.

Extensions to multi-agent systems. Recent work falls into four families (Table 1). None of them scores *individual sentences* while following influence *through other agents*, measures that influence by intervention, and validates it against a ground truth. This is the gap we target.

2 Setting

Simplifying hypotheses. Agents interact *only through generated text*: one agent's output is part of another's input, with no tools and no shared memory. Configurations are small: $N \leq 4$

Family	Examples	What it gives / what it misses
Attribute the last agents' input	ContextCite, Interpreto	Fine-grained (sentence, word), but only what the last agents read; influence that travelled through other agents is invisible.
Credit whole agents	Agents that Matter [8], Shapley-Coop [6]	Remove an agent and re-run (or let agents estimate it): one number per agent, no sentence-level explanation.
LLM judge of failures	Who&When [12] and follow-ups	A judge reads the log and names the faulty agent and step: an opinion, not a measurement; failures only.
Behavioural statistics	Debate or Vote [2], Not All Flips [5]	Show conformity and collapse on average; no explanation of a specific decision.

Table 1: Existing approaches to explaining LLM multi-agent systems.

agents and at most 3 rounds of discussion. Agents are open-weight models without a reasoning mode, so that we can read their probabilities and activations. Every task is multiple choice.

The discussion. N agents receive the same question. Some may also hold *private facts* that only they see. At round 0, each agent answers alone. At each later round, each agent reads some of the previous messages and writes a new one: a short reasoning, an answer and a confidence. At the end, the answers are aggregated by majority vote, or a designated agent decides (Figure 1). The *protocol* fixes two things: *who reads whom* (everyone, a chain, a star, ...), and *how much history* each agent sees (the whole discussion, or only the previous round). Messages are generated with fixed random seeds, so a discussion can be replayed exactly.

What we explain. We read from the logits the probability that each agent gives to each option, and define the support for the observed decision $\hat{y}$ as

$$v = \frac{1}{N} \sum_{i=1}^{N} p_i(\hat{y}),$$

where p_i is the final answer distribution of agent i. Explaining the decision means explaining v.

Units. Each message is split into sentences. The *units* are all sentences of all messages, plus the private facts, and we want a score for each unit. A unit x *feeds* a unit w if x was in the input from which w was written. These links form a graph that goes forward in time, the *influence graph*.

Baselines. We compare the discussion with a single agent, with self-consistency at the same compute budget, and with the *round-0 vote*, i.e. a majority vote before any discussion.

3 Measuring influence: frozen vs. re-run deletion

We measure the influence of a unit by deleting it and observing the change in v. The key question is what happens to the *rest of the discussion* when we delete it.

Definition 1 (Two ways to delete a unit)**.**

- **Frozen deletion.** Remove the unit u from every context where it appears, keep every other message exactly as it was written, and re-read the final answers. This gives $v^{\mathrm{F}}(u)$.
- **Re-run deletion.** Remove u, then re-generate every later message with the same seeds, and read the final answers. This gives $v^{\mathrm{R}}(u)$.

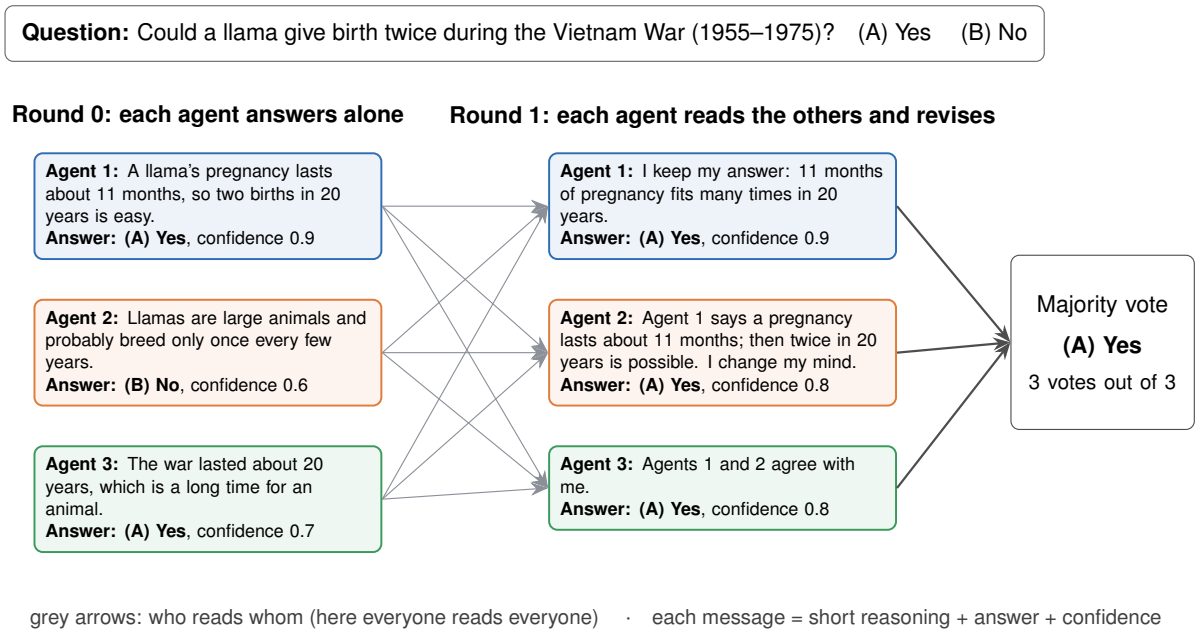


Figure 1: A complete discussion on a real example (StrategyQA-style question). Agent 2 first answers No, then reads Agent 1's argument (a llama's pregnancy lasts about 11 months) and switches to Yes. The decision is Yes, 3 votes out of 3.

The *direct*, *total* and *indirect* effects of u are

$$\mathrm{DE}(u) = v - v^{\mathrm{F}}(u), \qquad \mathrm{TE}(u) = v - v^{\mathrm{R}}(u), \qquad \mathrm{IE}(u) = \mathrm{TE}(u) - \mathrm{DE}(u).$$

The total effect answers "what did u cause?" and serves as our ground truth. It is expensive: one new discussion per unit. The indirect effect is the influence that travelled *through other agents*. Single-model methods (ContextCite, Interpreto) applied to the final agents only measure direct effects.

Proposition 1 (Attribution laundering)**.** *If the final agents never read a unit u, then* $\mathrm{DE}(u) = 0$*, and every method that only looks at the final agents' inputs gives u a score of* 0*, however large* $\mathrm{TE}(u)$ *is.*

Proof. Under frozen deletion all other messages are kept, and the final agents' inputs do not contain u. Their inputs, and therefore their answers, are the same with and without u. □

Figure 2 illustrates this on a chain where C reads only B. When the final agents read the whole history, the direct effect is no longer zero, but it is still *underestimated*: another agent's restatement of u stays in the context (Table 2).

	Final agents read the whole history	**Final agents read only part of it**
Frozen deletion (state of the art)	underestimated (restatement still present)	0 (Proposition 1)
Re-run deletion (ground truth)	true total effect	true total effect

Table 2: What each deletion measures for a key sentence that another agent restated.

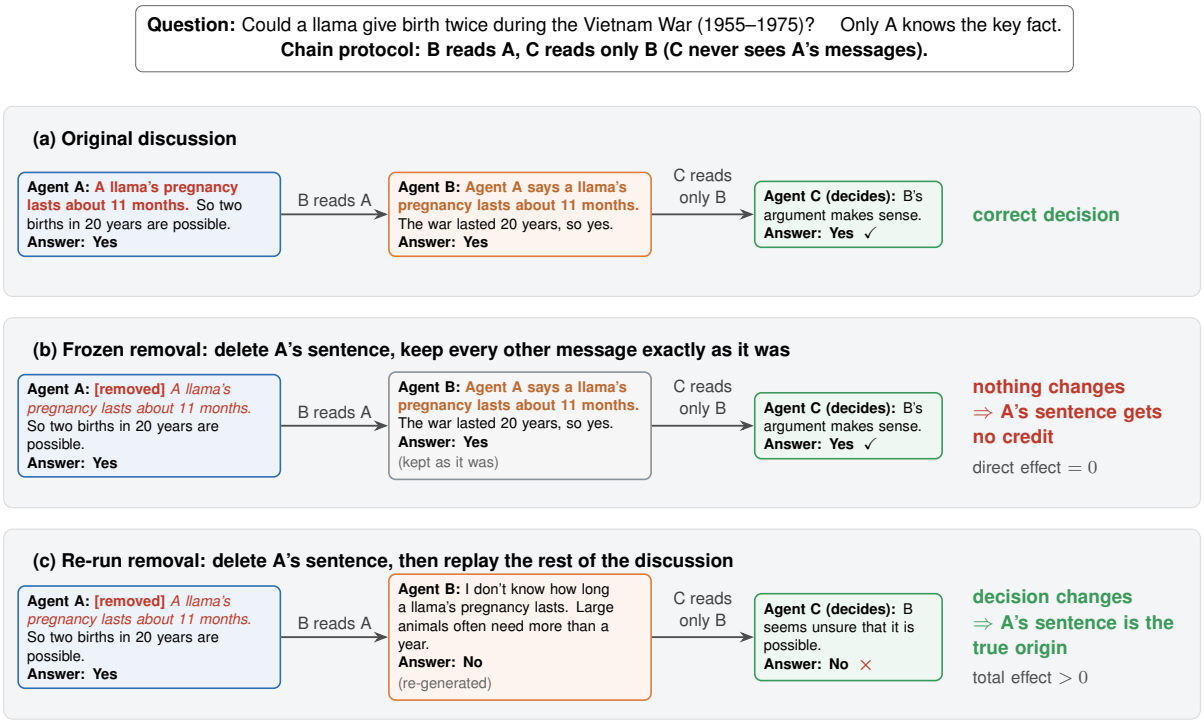


Figure 2: Attribution laundering on a chain (B reads A, C reads only B). (b) Frozen deletion of A's key sentence changes nothing for C, so DE $= 0$. (c) Re-run deletion: B no longer has the fact, and the decision flips, so TE > 0.

4 Proposed estimator: propagating local attributions

We want to approach TE without re-running the discussion. The discussion is generated *once*. Afterwards we only *re-read* it: we give a model a (possibly reduced) context together with the text it actually wrote, and read the probability it assigns to that text. No new text is generated.

Two local measurements.

- **Local shares.** For each unit w, we apply ContextCite to the agent that wrote w: we delete random subsets of its inputs and re-read w. This gives, for each input x, the *share* $a_{x\to w} \in [0, 1]$ of w that comes from x. For example, B's sentence "Agent A says a llama's pregnancy lasts about 11 months" comes 90% from A's sentence.
- **Direct effects.** We apply ContextCite to the final agents, and obtain a direct effect b_x for every unit they read ($b_x = 0$ for the others). This step alone is the state of the art.

Propagation. Going backward from the last round to round 0, the estimated total effect of a unit is

$$\widehat{\mathrm{TE}}(x) = b_x + \sum_{w \text{ fed by } x} a_{x\to w}\, \widehat{\mathrm{TE}}(w). \tag{1}$$

In words, the influence of a unit is its direct influence plus the influence of every unit it caused, weighted by how much it caused it: we multiply along each path and add up the paths (Figure 3). This is exact if influences combine linearly. It is the path-tracing rule of linear models [11], which we will state and prove precisely in the paper.

The main risk: redundancy. Real agents are not linear. When the same information reaches the decision twice, deleting one copy at a time makes each copy look useless. Deleting random

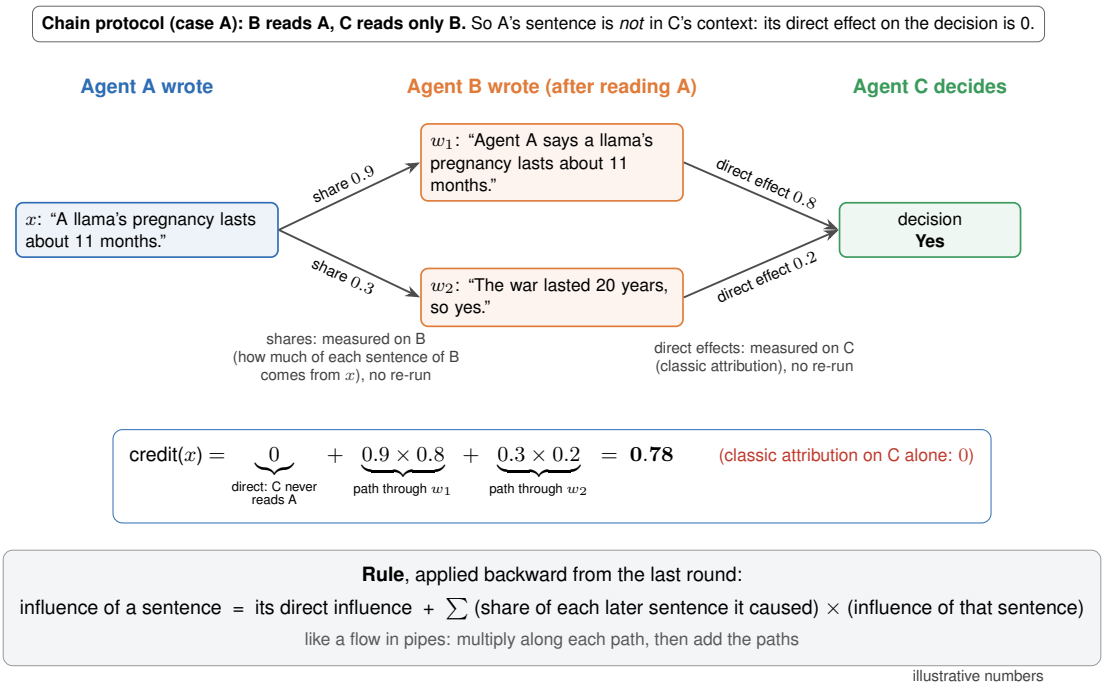


Figure 3: The estimator on the chain of Figure 2. Classic attribution on C gives A's sentence 0; propagation gives $0 + 0.9 \times 0.8 + 0.3 \times 0.2 = 0.78$. Numbers are illustrative.

Algorithm 1 What we run for one question

1: Run the discussion once with fixed seeds; store prompts, messages (split into units) and final probabilities. ▷ only generation
2: **for** each message **do**
3: Delete random subsets of its inputs (≈ 32 times), re-read the message, fit the local shares.
4: **end for**
5: Delete random subsets of the final agents' inputs (≈ 32 times), re-read the answers, fit the direct effects.
6: Propagate backward with Equation (1). ▷ no model call
7: *Validation only, on a subset:* compute TE by re-run deletion and compare.

subsets, as ContextCite does, sometimes removes both copies together, and the linear fit then shares the effect between them (Figure 4). How well this works in practice is the first thing we will measure.

Cost. Take one question with 3 agents, 3 rounds and about 45 units. Algorithm 1 needs about 320 re-readings and no generation. Re-run deletion needs up to 6 new messages per unit, i.e. about 270 generated messages. Measuring the actual speed-up is part of the evaluation.

5 Evaluation

Ground truth by construction. On real questions, the model may already know the answer, and information may arrive through several routes. We therefore build *synthetic tasks* with invented entities whose facts are distributed among agents (Figure 5). Because the names are invented, the answer can only come from the facts. We know by construction which facts determine the answer. Because the names are unique, we also detect exactly which sentences restate them. Variants include 1 to 3 hops, one agent holding all facts, misleading shared facts (hidden profile

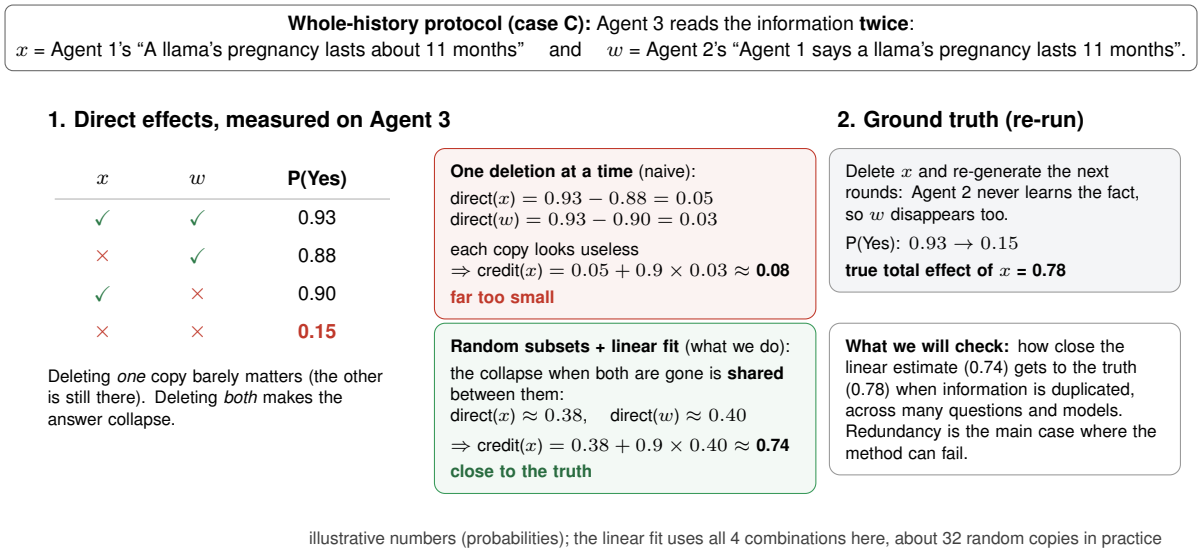


Figure 4: Redundancy when the final agent reads the whole history. One deletion at a time gives A's sentence a credit of ≈ 0.08. Random subsets with a linear fit give ≈ 0.74, close to the ground truth 0.78. Numbers are illustrative.

[7]), and planted false facts.

Metrics. We compare three attributions: frozen methods, our estimator, and an LLM judge. Each is measured against the ground truth on three criteria:

- **agreement**: rank correlation with TE over units;
- **origin share**: the fraction of the credit given to the true source facts rather than to their restatements. Laundering means an origin share close to 0;
- **cost**: number of model calls and generated tokens.

Explaining the multi-agent gain. On real benchmarks (StrategyQA, MMLU-Pro), we split the gain over a single agent into a *voting* part (round-0 vote vs. single agent) and a *discussion* part (final decision vs. round-0 vote). Each change of answer is then attributed, with our estimator, to private facts (information transfer), to the reasoning of other agents (persuasion), or only to their stated answers (conformity).

6 Hypotheses and next steps

H1 Laundering. On chain tasks, frozen methods give the true source almost no credit; our estimator recovers it.

H2 Cost. Our estimator agrees with TE almost as well as re-run-based methods, at a small fraction of their cost.

H3 Sources of the gain. On standard benchmarks the gain comes mostly from voting. When information is distributed among agents, it comes from the discussion and is mostly indirect.

H4 Mechanism (white-box). Agents that switch answer by conformity still represent their previous answer internally more often than agents persuaded by reasoning (linear probes on activations, cf. [9]).

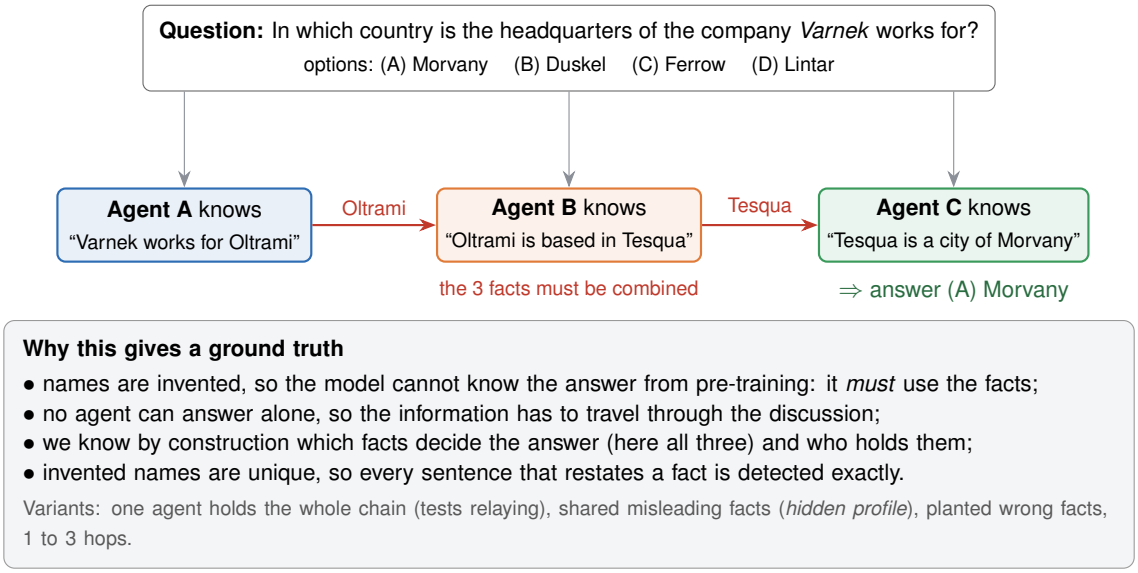


Figure 5: A synthetic task: each agent knows one invented fact, and the three must be combined to answer.

Next steps. (1) Validate this formalization. (2) Build the testbed: 3 agents, debate protocol, Qwen2.5-7B-Instruct, synthetic tasks and StrategyQA, with logging of messages, probabilities and activations. (3) Measure behaviour: single agent vs. vote vs. debate, and the disagreement rate. (4) Test H1 on chain tasks, a target for a workshop paper in early 2027. (5) Implement the estimator and scale to three model families (H2, H3), then the white-box study (H4), targeting NeurIPS 2027.

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